# CAPSTONE PROJECT

## PROJECT TITLE:-
## AI-Based Olympic Medal Count Prediction System


PRESENTED BY

STUDENT NAME:- UDDIP BISHT

COLLEGE NAME:- Panipat Institute of Enginnering & Technology

DEPARTMENT:- B.TECH (CSE)

EMAIL ID:- uddipbisht2004@gmail.com

AICTE STUDENT ID:- STU668e4423c86ed1720599587


# OUTLINE

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Output Image)
- Conclusion
- Future Scope
- References

# PROBLEM STATEMENT

•The Olympic Games witness participation from countries across the world, aiming to secure medals in various sports.

•Predicting medal outcomes can help in resource allocation, training focus, and performance benchmarking.

•This project addresses the challenge of building a machine learning model to predict the number of medals a country may win based on historical Olympic data.

•The objective is to use available athlete and event-level data to generate country-level medal count predictions.

# PROPOSED SOLUTION

- A supervised machine learning model is proposed for medal prediction.
- The model utilizes features such as country (NOC), athlete statistics, and number of events.
- Aggregation of athlete-level data to team (country) level helps create the training dataset.
- The solution involves preprocessing data, training a regression model, and validating results with real data.

# SYSTEM APPROACH

**Technology Stack Used:**

- **Language:** Python
- **Libraries:** Pandas, NumPy, scikit-learn, matplotlib, seaborn
- **IDE/Platform:** Jupyter Notebook
- **Data Source:** teams.csv + cleaned athlete dataset from Olympic Games

# ALGORITHM & DEPLOYMENT

- **Algorithm Used:**
- Linear Regression Model (or similar regression-based ML model)
- Grouping by country (NOC) and year
- Feature engineering: calculating average age, height, weight, number of events, etc.
- **Deployment:**
- The model was trained and evaluated within the notebook.
- Can be deployed via a web interface or integrated into sports analytics platforms in the future.

# RESULT

- The model successfully predicts medal counts with decent accuracy.
- The result is visualized as shown below:

In [60]:
```
test["predictions"] = predictions
```

In [61]:
```
test[test["team"] == "USA"]
```

Out[61]:

| | team | country | year | athletes | age | prev_medals | medals | predictions |
|---|---|---|---|---|---|---|---|---|
| 2053 | USA | United States | 2012 | 689 | 26.7 | 317.0 | 248 | 285.210121 |
| 2054 | USA | United States | 2016 | 719 | 26.4 | 248.0 | 264 | 235.568076 |

In [62]:
```
test[test["team"] == "IND"]
```

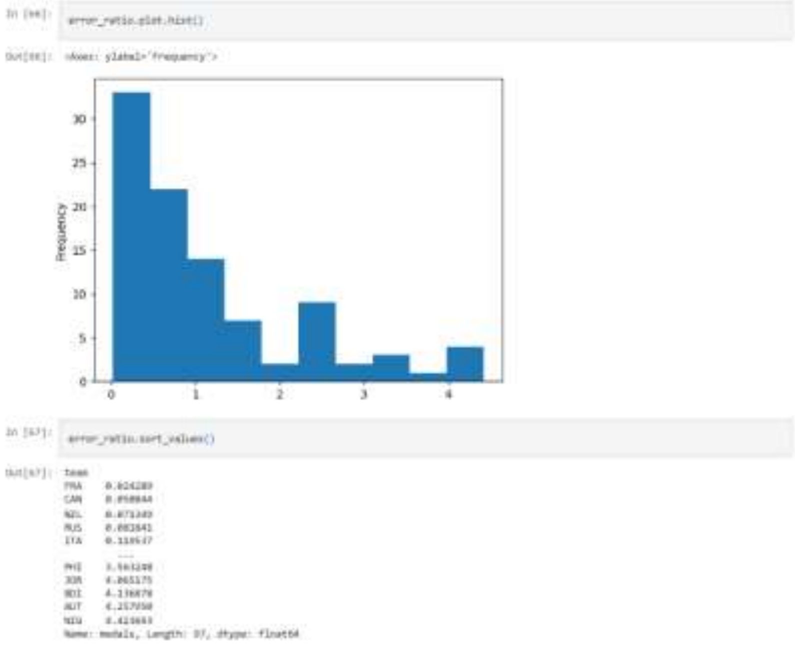
Out[62]:

| | team | country | year | athletes | age | prev_medals | medals | predictions |
|---|---|---|---|---|---|---|---|---|
| 907 | IND | India | 2012 | 95 | 26.0 | 3.0 | 6 | 6.921667 |
| 908 | IND | India | 2016 | 130 | 26.1 | 6.0 | 2 | 11.683176 |

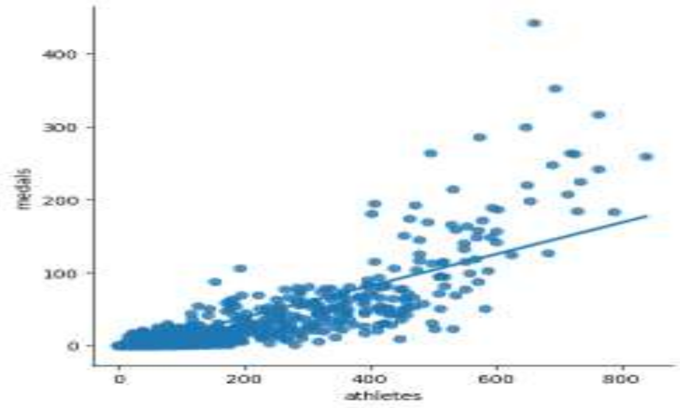
In [63]:
```
errors = (test["medals"] - predictions).abs()
```

In [64]:
```
error_by_team = errors.groupby(test["team"]).mean()
medals_by_team = test["medals"].groupby(test["team"]).mean()
error_ratio =  error_by_team / medals_by_team
```

In [65]:
```
import numpy as np
error_ratio = error_ratio[np.isfinite(error_ratio)]
```



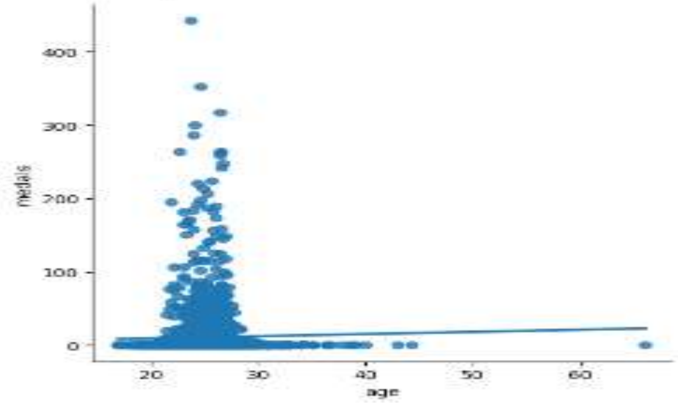
Out[39]: <seaborn.axisgrid.FacetGrid at 0x1d4e43d96a0>




In [40]:

```
sns.lmplot(x='age', y='medals', data=teams, fit_reg=True, ci=None)
```

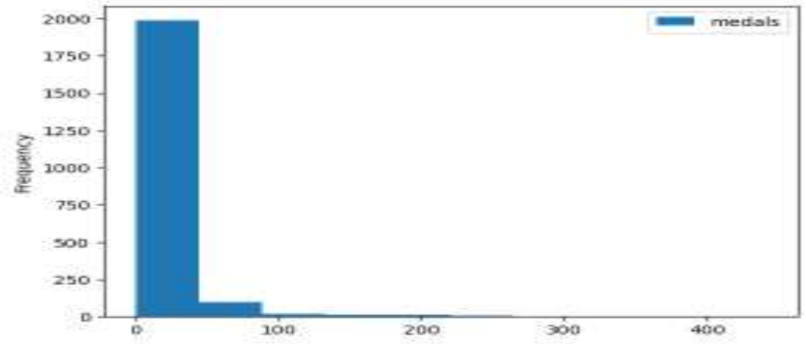
Out[40]: <seaborn.axisgrid.FacetGrid at 0x1d4e3254910>




In [41]:

```
teams.plot.hist(y="medals")
```

Out[41]: <Axes: ylabel='Frequency'>



# CONCLUSION

- The project demonstrates that historical athlete and event data can be used effectively to predict national Olympic performance.
- It provides a framework for federations to analyze and optimize their sports strategies.
- Further tuning and more sophisticated models (e.g., ensemble methods) may improve accuracy.

# FUTURE SCOPE

- Integration of deep learning models and neural networks for complex pattern recognition.
- Use of real-time performance data (e.g., athlete rankings, recent wins) to improve prediction.
- Extension to individual athlete-level medal predictions.
- Deployment as a dashboard or app for sports analysts and federations.

# REFERENCES


•Kaggle Olympic History Dataset

•Python (pandas, scikit-learn documentation)

•Jupyter Notebooks

•IOC Official Website for historical data validation


GitHub Link: https://github.com/uddipbisht/AI-ML-1-Project.git

# Thank you